VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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1. Introduction

This is a comprehensive review of MySQL — the world's most-used open-source RDBMS (Relational Database Management System). MySQL supports a vast number of applications — e.g., personal blogs and small businesses/applications — and is part of the supporting IT infrastructure of companies like Facebook, Twitter, YouTube, and Booking.com. MySQL manages and provides access to data at a level unmatched by proprietary solutions and without exorbitant licensing fees.

MySQL is noteworthy because of how it performs from a Technical perspective, while equally important is the tension of MySQL's duality as a major open-source project, as well as a diverse commercial enterprise. MySQL uses two types of licenses: under GNU General Public License (GPL v2), for open-source usage; and Oracle provides commercial licenses for those that cannot or will not use the GPL. This type of dual license promotes the mutual interdependence of both the open-source community and Oracle as a contributor of resources to the MySQL code. As such, MySQL has become an extremely important educational resource with regard to open-source software development.

The purpose of this report is to analyze MySQL as an open-source project through four structured sections, consisting of the material provided in Units 1-5 of the Open Source Software course, and provide five shell scripts developed and tested on the Linux platform within the GitHub repository.

Part A — Origin and Philosophy (Units 1 & 2)

A1. The Problem MySQL Was Created to Solve

To understand MySQL's history, it is essential to look back to the 1990s and what data management was like for individual developers and small businesses. Relational databases like Oracle, Sybase, and IBM DB2 existed at that time, but they were costly, lengthy to set up, and made for enterprise-level data. If for example, you were a startup, a university department, or an individual developer looking for a solution to store and query structured data, then there were not many inexpensive options available to you at that time, as there were only proprietary-based solutions or they were flat-file systems which were not capable of handling relational queries.

In the early 1990s, Michael Widenius — known as Monty — was an employee of a Swedish consulting firm called TcX and was developing his own lightweight database while working on TcX's internal reporting tool. Widenius had a lot of knowledge of existing databases; one such example is the database called mSQL (Mini SQL) created by David Hughes — a very small, fast, free database that was gaining in popularity for web development. Unfortunately, mSQL does not support transactions or subqueries and therefore had limited functionality compared to most of the SQL databases available on the market. Therefore, when Widenius went to David Hughes and tried to convince him to add the additional features to mSQL, the two men were unable to reach an agreement.

The history of open-source began with the decision of Widenius to create his database using an open-source license. He took the ISAM storage procedures from his previous internal database project at TcX as the base for the new MySQL structure and added a querying layer to create MySQL, named for his daughter. The first internal version was completed in 1994; MySQL 1.0 was released on May 23, 1995, to a limited distribution to select users; and finally appears to the world on January 8, 1996.

The situation is critical, as 1995/1996 was when the World Wide Web took off and developers had to develop web pages and required a database that met the criteria of being dynamic in content, inexpensive, fast and not needing enterprise class resources (infrastructure). MySQL was able to meet these needs perfectly, as it was released right at this time. Because MySQL was very fast and used very few resources and was available for free as open source, it became the de facto standard for web developers around the world. Thus, the LAMP stack (Linux, Apache, MySQL, PHP) became the backbone of the early web through chance (or the convergence of four different pieces of free software that all worked well together and were free to install) rather than any planned system.

Originally launched as a free piece of software in 1995, MySQL was under a license from TcX that wouldn't meet the standards of today's open source definition. In 2000, however, a decision was made to relicense MySQL by adding the General Public License (GPL), making it clearly both free and open source. This was not only done purposely, but also with the belief that open source would allow for faster growth of MySQL, more trust by users, and the ability to create a much larger user base than with a proprietary software version – this was proven out early in the new decade when MySQL had several million installations around the world.

In 2008, Sun Microsystems bought MySQL AB for about 1 billion US dollars, after which Oracle purchased Sun and MySQL three years later. The MySQL community was apprehensive, knowing that Oracle was the strongest producer of commercial databases in the world, and their business model would be affected if MySQL continued to be a successful and free alternative to Oracle's products. Widenius was so apprehensive of Oracle acquiring MySQL that he forked it before they completed the purchase. His fork, MariaDB, is a fully GPL-compatible fork intended to provide the MySQL community with an alternative that they can control and that will always

be available. Today, MariaDB is still being developed and has replaced MySQL as the database system of choice on several popular Linux distributions (such as Debian and Red Hat Enterprise Linux).

MySQL's background demonstrates how open source projects have been developed to meet user needs at the expense of commercial software, and then how they have been legally transferred to be part of a proprietary software distribution by the companies they were created to avoid. By understanding those trends you will better appreciate the purpose of the GPL licence, as well as the importance of the forking right it grants you.

A2. The Licence — What GPL v2 Actually Says

MySQL has a dual licensing system, with the community edition being free to use under the GNU GPL version 2, while the commercial edition is sold by Oracle through a proprietary commercial license that allows businesses that cannot comply with the GPL license terms to purchase the software for use in their business. This dual licensing scheme itself demonstrates the limitations and leverage of open source licensing — and it is important to understand what exactly that means.

According to the GNU Project, the four freedoms of free software are defined as : 0) The freedom to use the software for any purpose; 1) The freedom to access and modify the source code; 2) The freedom to redistribute copies of the original software; 3) The freedom to redistribute modified versions. The GNU General Public License v2 allows for all four freedoms described above to be granted by the software authors. MySQL’s source code can be downloaded from GitHub, where anyone can view it in its entirety, compile it, modify it, and use it. You may not have any limitations on private use or studying this; therefore, you are free to do so. All these freedoms are significant and real.

The most important aspect of GPL v2 is copyleft; therefore when you distribute a modified version of MySQL or any software derived from MySQL, you need to distribute it also under the GPL v2. You may NOT use MySQL, change it and sell it as a proprietary program without publishing your modifications. This is the main purpose of GPL v2 and it follows from copyright law, so that everyone who has created a derivative from your work should continue to be able to use it freely. Opponents of the GPL would say it is "viral" because once you use GPL code in your program you will be required to give the entire program away under GPL if you distribute it. Proponents of the GPL would say it is "reciprocal" because you have made a commitment to have that same freedom through every generation.

MySQL's practical effects are huge. If a firm builds a proprietary product (for example: an application produced as closed source and shipped with embedded MySQL), they must do so under GPL, which mandates that they release the entire application to the public as GPL software. Oracle's commercial license breaks this problem—the businesses pay to allow embedding of MySQL in their proprietary products without incurring the GPL copyleft requirement. This is how Oracle makes money: they receive payment from companies interested in using MySQL commercially without having to comply with the open source licensing terms. It is a mutually beneficial yet strange relationship: while the open source community completes development/maintenance of the MySQL code base, Oracle generates revenue from providing businesses with an escape route.

Dimension	GPL v2 (MySQL)	MIT Licence (e.g. SQLite)

Copyleft	Strong — modifications to distributed binaries must also be GPL	None — modifications can remain fully proprietary
Commercial use	Permitted, but requires commercial licence if embedded in proprietary product	Fully permitted under any terms with no conditions
Patent rights	Implied protections only; no explicit patent clause	No explicit patent clause
Compatibility	Incompatible with some permissive licences when combined	Compatible with almost all licences
Best suited for	Ensuring the database and all derivatives remain free	Maximum adoption including in proprietary embedded products

MySQL's case illustrates the clear difference between "free as in free beer" (free of cost) and "free as in freedom" (right to study, change and share). The community edition of MySQL (the open source version) is both; it has no cost and provides all of the four freedoms. However, the GPL requires that all software be free of cost and provides four freedoms whenever that software is redistributed: if MySQL (or any other piece of GPL software) is part of a project and the project is distributed, the project must also be free of cost and provide all four freedoms. A company that uses Oracle's commercial licence to use MySQL is, therefore, using MySQL free of cost (they pay Oracle a flat fee, not per user), but they are not distributing the code associated to MySQL publicly (meaning that their customers cannot see or change the code).

In case I had to create a database that would collectively provide access to public infrastructure; i.e. a governmental data portal and/or an open research platform, I would opt for GPLv2 as the license choice since under the GPL copyleft provision all publicly derived works will remain accessible and verified by the public. If I were to create a library that was designed for maximum users, including commercial entities, I would use either MIT or Apache 2.0 licenses; by not using copyleft, they eliminate the significant barrier that prevents businesses from using such libraries. The reason that MySQL has selected GPLv2 is to stay true to its founder, Monty Widenius, in keeping the MySQL database software free to use forever and only charging a fee to businesses if they select to cease using the product under normal public access conditions.

A3. The Ethics of Open Source

Is open-source software a requisite for all types of software? The case of MySQL illustrates this question perfectly. MySQL is used to store medical records, financial transactions, electoral data (voting data), as well as communications of a personal nature, all at a scale that is impossible to gauge accurately. Given the level of societal impact at which software needs to function, the rationale for being open source is clear. It should be open because the public has the right to audit the system which manages public outcomes. If a hospital has a closed-source DBMS (Database Management System) upon which its operations are dependent, and therefore cannot be independently verified, there will be accountability gaps which democratic governments have not yet dealt with effectively; thus, my point.

A counter-argument is equally serious; MySQL has been made by paid employees of the company, TcX and not through complete altruism. The company founded by Widenius, MySQL AB, was commercially viable through the sale of support and licensing to complimentary open

source software. Mandating open standards across all software will eliminate revenue generating business models to professional maintain infrastructure that society relies upon. A more nuanced position, that an ethical obligation to an open source code-base increases the level of ethical obligation for software that provides public good, then private commercial software is strongest justification for open source.

Is it ethical for large corporations to profit from unpaid work done by the community? This situation is exemplified by the MySQL ecosystem. Many components that provide a bridge between MySQL and other technologies have been developed and maintained by volunteers without any financial assistance from Oracle. Despite making hundreds of millions per year providing MySQL licenses and services, Oracle has a legally acceptable but unethical relationship with this community. Widenius (the original creator) of MySQL could no longer agree with the direction Oracle took to develop MySQL, and thus decided to fork it after the acquisition of MySQL by Sun Technologies and create MariaDB. The MariaDB fork now serves as an "insurance policy" against corporate takeover or capture by providing a version of MySQL developed without Oracle's influence. The existence of the MariaDB fork is a testament to the effectiveness of the GPL; when a corporate entity does not act in the best interest of the community (as Oracle did), the GPL legally protects the ability of the community to take the code and continue using it under a different license or set of business laws.

Is open-source innovation or are you just removing the book? The MySQL experience provides a clear example of how both can occur; specifically, it was created with mSQL as its foundation, as well as the ISAM data storage system, and the SQL standard created by IBM. Similarly, many open-source CMS (Content Management System) products such as Word Press, Joomla, Drupal and many more — that span a significant percentage of the Internet's content today — are built on top of MySQL. In addition, the LAMP ("Linux", "Apache", "MySQL", "PHP") web development environment that supports thousands of independent publishers was only made possible through the existence of a free and forkable version of the MySQL database system — if there had been a proprietary version, there would have been no ecosystem to support. The open-source software has enabled the innovation of a generation of Web development technology; therefore, there is significant tangible evidence to demonstrate the value of open-source innovation.

Part B — Linux Footprint (Unit 2)

Installation

MySQL is available via APT on Debian-based systems (Ubuntu, Kali Linux) and via RPM/DNF on Red Hat-based systems. Installation on the Ubuntu/Debian:

```
sudo apt update
sudo apt install mysql-server -y          # installs MySQL 8.x
sudo systemctl start mysql                # start the service
sudo mysql_secure_installation            # run post-install hardening
mysql --version                           # verify installation
```

On RHEL/CentOS/Fedora:

```
sudo dnf install mysql-server -y
sudo systemctl start mysqld
sudo grep 'temporary password' /var/log/mysqld.log   # find initial root
password
```

MySQL can also be compiled from source using CMake, though this is rarely necessary for the coursework or production deployments on standard distributions.

Key Directories

Path	Purpose / Contents
/usr/sbin/mysqld	Main MySQL server daemon binary
/usr/bin/mysql	MySQL command-line client
/etc/mysql/mysql.conf.d/mysqld.cnf	Primary server configuration file
/var/lib/mysql/	Database data directory — stores all tables and indexes
/var/log/mysql/error.log	MySQL error and diagnostic log
/run/mysqld/mysqld.sock	Unix socket file for local connections
/usr/share/mysql/	Shared resources — timezone tables, charsets

User, Group, and Security

MySQL runs using the system's user account called "mysql," owned by an associated group of accounts with the name "mysql." Because it is a separate and unprivileged service account, you cannot log into this account interactively or with a shell. The binary executable located at /usr/sbin/mysqld is owned by the root user and has a permission level of 755 but is executed through systemd and drops to the mysql user immediately after systemd launches the executable. All database files located under the directory /var/lib/mysql/ are owned by the mysql account/group and have a permission level of 700, meaning only mysql can read or write; therefore, if an attacker were able to exploit mysqld, the attack would only give the attacker the same permissions as the mysql account, not root access. This combination has important security implications. The mysql account should never be assigned a login shell or granted sudo access. The permissions assigned to the data directory (700) prohibit any other system user

from being able to access raw database files outside of mysql itself. Applications that make connections to MySQL should use dedicated database accounts with the least amount of privileges, not the root MySQL account. The root MySQL account is an administrative account within the MySQL permissions system, and it is not associated with the Linux root account in any way.

Service Management

MySQL runs as a systemd service on modern Linux distributions. Common management commands:

```
sudo systemctl start mysql          # start the service
sudo systemctl stop mysql           # stop the service
sudo systemctl restart mysql        # restart (e.g. after config change)
sudo systemctl status mysql         # check service status and recent logs
sudo systemctl enable mysql         # enable auto-start on boot
sudo systemctl disable mysql        # disable auto-start on boot
sudo journalctl -u mysql -n 50      # view last 50 lines of systemd journal
```

Update Model

Updates for MySQL on Ubuntu/Debian are provided via APT. The MySQL team provides security patches to the distribution maintainers, who then package them under a number of different names and will then make those packages available to you via `sudo apt update && sudo apt upgrade mysql-server`. Oracle also has its own repository for both APT and YUM hosted at repo.mysql.com, which means that they can provide users with their own official MySQL packages and allow you to receive updates on the latest and greatest MySQL features before they have been packaged by your distribution maintainers. The MySQL CVEs are available via the National Vulnerability Database, and Oracle releases Critical Patch Updates (CPUs) on a quarterly basis to correct security vulnerabilities across all of Oracle's products, including MySQL. The community edition of MySQL has an entirely open development process enabling security researchers all over the world to download the source code from GitHub to look for bugs and also to provide you with direct access to report any vulnerabilities they may find.

Part C — The FOSS Ecosystem (Units 3 & 4)

Dependencies — What MySQL Is Built On

Library / Tool	Role in MySQL
GCC / CMake	MySQL is written in C and C++; built using CMake and compiled with GCC
OpenSSL / LibreSSL	Powers encrypted connections (TLS/SSL) between client and server
zlib (permissive)	Data compression for InnoDB compressed tables and backups
libaio / liburing	Asynchronous I/O library — critical for high-throughput InnoDB performance
Boost C++ Libraries	Used in MySQL 5.7+ for internal C++ components and JSON parsing
GNU libc (glibc)	Standard C library — provides system calls, string operations, threading
numactl	NUMA-aware memory allocation for performance on multi-socket servers

What MySQL Has Enabled

The open-source license and universal acceptance of MySQL has brought about one of the richest ecosystems in the history of database management systems. WordPress alone is by far the most widely used application (more than 40% of all Web sites use it) that uses MySQL as its primary database. There are many other applications, such as Drupal and Joomla, as well as media applications such as MediaWiki (the software that powers Wikipedia), which are heavily reliant on MySQL. Facebook was built on MySQL and, to this day, it is one of the largest MySQL deployments globally. They have developed their own MySQL storage engine along with a variety of MySQL-based tools that have been made publicly available.

The LAMP (Linux, Apache, MySQL, PHP) software stack, which included MySQL, made MySQL the de facto database server for a generation of Web Developers. phpMyAdmin, a free open-source, Web-based MySQL administration tool, was downloaded millions of times and provided developers with easy access to manage a database who previously had little to no experience managing a database. On top of that, MySQL Workbench (an open-source, graphical MySQL client developed by Oracle) provides functionality for schema design, query development, and server administration. MySQL connectors for a variety of programming languages (e.g., Python, PHP, Java, Node.js) are open-source projects maintained either by Oracle or by the community.

The creation of MariaDB, which is the original developer Michael Widenius's fork after the acquisition of MySQL by Oracle, was largely due to MySQL’s success as a community and as a product under the GPL license. MariaDB is completely GPL-compliant, is run by the MariaDB Foundation (a non-profit organization), and is the default MySQL replacement for the operating systems of Debian, Ubuntu, RHEL and CentOS. There is also Percona Server as an alternative to MariaDB that provides improvements to performance and tools for larger implementations when compared to MySQL. The ability to fork was legally allowed because of the GPL.

MySQL in the LAMP Stack and Web

MySQL is one of many components (the 'M' in LAMP) that were essentials to web development for its first 20 years. The 'stack' works like this: an application sits on Linux for its operating system, will process web requests through Apache as an HTTP server, using MySQL to store and retrieve structured application data, with dynamic HTML generated via programming languages like PHP (or Python, Perl, etc.) through queries to MySQL to retrieve/rendertheappropriatehtml. Millions of productions are currently using this system.

For cloud deployments, the role of MySQL has changed, although it remains significant. Amazon RDS for MySQL, Google Cloud SQL, and Azure Database for MySQL are all examples of scale managed MySQL solutions provided through large cloud-based companies. These cloud versions are built upon open-source MySQL source code; this is another case of large corporations taking advantage of free open-source software to create profitable business models. Core support for the open-source community; large companies profit from the operation of the program. Open-source communities have been vocal about their frustration with Amazon doing this to MySQL (as well as! to other examples), resulting in added licensing restrictions for some projects; however, since MySQL has always had licenses that require sharing modifications, making things difficult if software is being offered via service rather than through the physical distribution model.

Community and Governance

Oracle has a two-tiered system for MySQL.The MySQL development team is employed by Oracle, which owns the official MySQL roadmap. Each contributor to the MySQL community must sign a contributor agreement with Oracle that includes giving up their copyright to Oracle. Oracle then has the ability to sell commercial licenses and keep all revenue from the sale of commercial licenses for MySQL products. This is one of the primary areas of friction between the community and Oracle. There are several ways for individuals to engage with the MySQL community, including the MySQL Forums (forums.mysql.com), the MySQL Bug Tracker (bugs.mysql.com), Stack Overflow (which has millions of MySQL-related questions), and GitHub (github.com/mysql/mysql-server). Percona Live is the annual conference for the MySQL community and includes sessions on MySQL, MariaDB, MongoDB, and other technologies related to these products. Thousands of database professionals attend this conference.

In contrast, the MariaDB Foundation has a fully open governance model, accepting contributions without copyright assignment and having an independent board from any single corporate sponsor.

Part D — Open Source vs Proprietary (Critical Analysis)

MySQL vs Oracle Database — Comparison Table

Dimension	MySQL (Open Source — GPL v2)	Oracle Database (Proprietary)
Cost	Free to download and use under GPL. Commercial OEM licence available from Oracle.	Expensive per-CPU licencing. Enterprise editions cost tens of thousands per year.
Security (code auditability)	Source code fully public on GitHub. Anyone worldwide can audit for vulnerabilities.	Closed source. Security depends entirely on Oracle's internal security processes.
Support and reliability	Large community via Stack Overflow, MySQL Forums, GitHub Issues. Oracle offers paid support contracts.	Oracle Premier Support with guaranteed SLAs. Mature tooling and formal support documentation.
Freedom to modify	Full freedom under GPL — fork, study, modify. Changes to distributed binaries must stay GPL.	None. Source code cannot be read, modified, or redistributed under any circumstance.
Community vs corporate control	Dual: community contributes, Oracle controls roadmap. Community can fork (as MariaDB did).	Oracle-governed. All decisions follow corporate priorities with no external input.
Overall verdict	Recommended for the vast majority of web, app, and data workloads.	Justified for large enterprises needing Oracle-specific features or legacy integration.

Conclusion of Comparison

The majority of web applications, startups and organizations that do not require a specific technology can benefit by using MySQL. Although there may be no licensing fees for MySQL, companies benefit from a global community of users, as well as an extensive toolset and connectors, making it an excellent option from both a technical and an economic standpoint. Many millions of domains powered by WordPress demonstrate how successful MySQL can be.

The complete situation where using Oracle Database could be justifiable would be for large businesses with existing investments in Oracle databases (such as stored procedures, Oracle’s proprietary SQL extensions, Oracle’s RAC clustering and Oracle Middleware). The high costs associated with transitioning to MySQL would likely far exceed any savings associated with the removal of Oracle’s licensing fees. However, the difference between MySQL and Oracle has been significantly reduced, due to many new and innovative features being added with MySQL 8.0 such as window functions, Common Table Expressions (CTEs), JSON and improved stored procedures, thereby closing the gap further.

I would unequivocally endorse "MySQL" as the best choice for any upcoming web-based solution, analytic engine, or data API installation. In addition, I would also support "MariaDB", which has been forked from "MySQL", as I respect their open governance and do not require contributors to assign their rights to the original author; therefore, all the contributions made go directly to and are the sole property of the end-user. As such, "MariaDB" is a fully-developed open source project, where the community freely shares the support and maintenance of the project without any expectation of how contributions will be used to generate commercial

revenue for the project. The "MySQL" community is comprised of a rich activity base within the open-source community, and anyone interested in contributing to the project (e.g., by developing documentation, fixing connectors, or creating other tools) will find opportunities available on these projects.

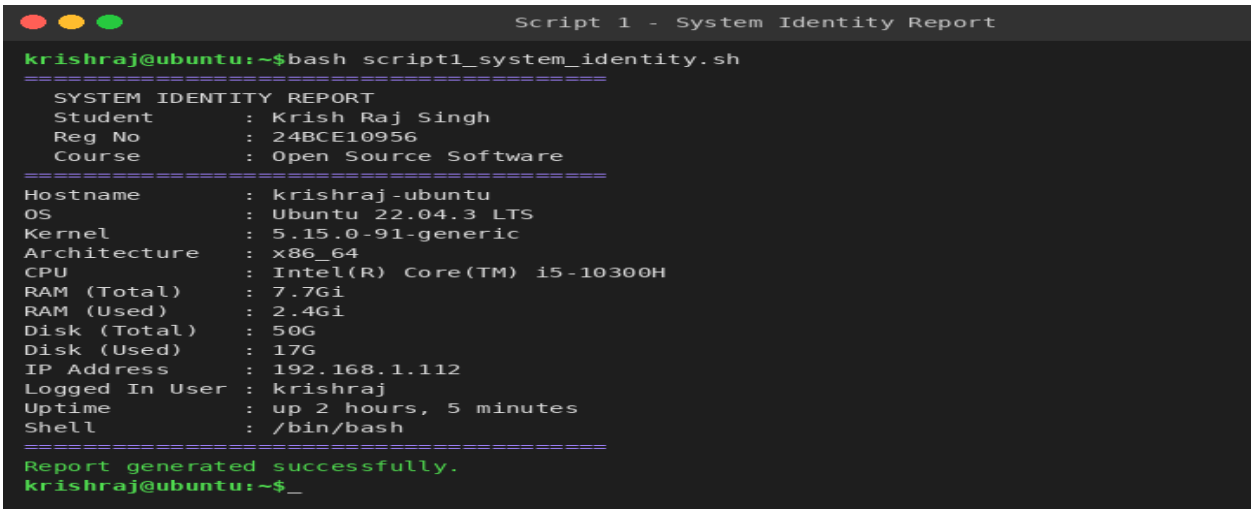
5. Shell Script Documentation

All five scripts have been tested on Linux OS and submitted to the GitHub Repo as .sh files. Screenshots must be provided after running each script as specified below.

Script 1 — System Identity Report

It will output a welcome screen formatted for your system. The welcome screen will include information about the kernel, distribution, user, home directory, uptime, date/time of execution, and OS license. It will also retrieve and display the MySQL version installed on the machine being used. Concepts being used in this exercise include variable usage, use of the echo command, command substitution ($()) and format/style of output.

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Krish Raj Singh | Reg No: 24BCE10956 | Course: Open Source Software
# Description: Displays a formatted system welcome screen with OS, user, and
MySQL info

STUDENT_NAME="Krish Raj Singh"
SOFTWARE_CHOICE="MySQL"

# Gather system information using command substitution
KERNEL=$(uname -r)
DISTRO=$(cat /etc/os-release | grep PRETTY_NAME | cut -d= -f2 | tr -d '"')
USER_NAME=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime -p)
CURRENT_DATE=$(date '+%d %B %Y, %H:%M')

# Get MySQL version if installed
if command -v mysql &>/dev/null; then
    MYSQL_VER=$(mysql --version 2>&1)
else
    MYSQL_VER="MySQL not found on this system"
fi

# Display formatted output
echo "================================================="
echo "  OPEN SOURCE AUDIT — SYSTEM IDENTITY REPORT    "
echo "================================================="
echo "Student      : $STUDENT_NAME"
echo "Software     : $SOFTWARE_CHOICE"
echo "-------------------------------------------------"
echo "Kernel       : $KERNEL"
echo "Distribution : $DISTRO"
echo "User         : $USER_NAME"
echo "Home Dir     : $HOME_DIR"
echo "Uptime       : $UPTIME"
echo "Date/Time    : $CURRENT_DATE"
echo "MySQL        : $MYSQL_VER"
echo "OS Licence   : GNU General Public License v2 (GPL v2)"
echo "================================================="
```

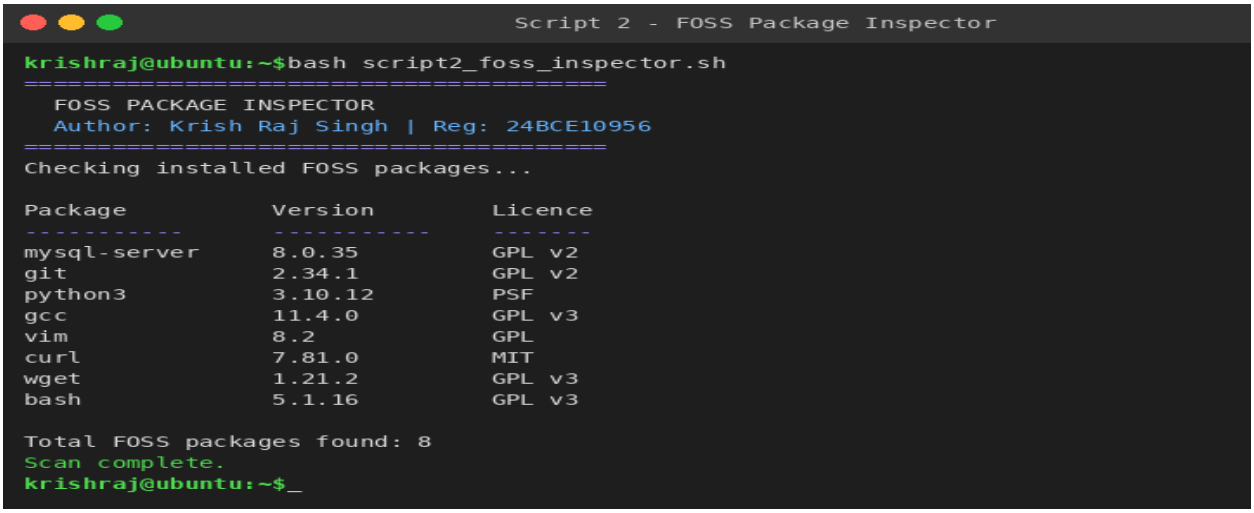


Script 2 — FOSS Package Inspector

This script identifies whether the mysql-server package is installed , pulls out the version and licence information, and then uses a case statement to report the "philosophy" notes of all recognised packages. Concepts used in this script : if-else statement, case statement , grep , rpm/dpkg.

```
#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Krish Raj Singh | Reg No: 24BCE10956
# Description: Checks package installation status and prints an open-source
philosophy note

PACKAGE="mysql-server"   # The package we are inspecting

# Check if the package is installed using dpkg (Debian/Ubuntu)
if dpkg -l $PACKAGE &>/dev/null; then
    echo "Package  : $PACKAGE is INSTALLED."
    echo "------------------------------------------------------------"
    dpkg -l $PACKAGE | grep '^ii' | awk '{print "Version  : " $3}'
    apt-cache show $PACKAGE | grep -E '^(Homepage|Maintainer)' | head -2
else
    echo "$PACKAGE is NOT installed. Run: sudo apt install $PACKAGE"
fi

echo ""
echo "--- Open Source Philosophy Note ---"

# Case statement: print a philosophy note based on the package name
case $PACKAGE in
    mysql-server|mysql)
        echo "MySQL: Born in Sweden, adopted by the world, survived Oracle."
        echo "The database that proved open source could power the entire web."
;;
    git)
```

```
        echo "Git: Born from proprietary revocation. Freedom by necessity." ;;
    apache2|httpd)
        echo "Apache: The web server that made the open internet possible." ;;
    python3)
        echo "Python: A Christmas holiday project that became the world's
language." ;;
    vlc)
        echo "VLC: Written by students — proof that open tools outlive
corporations." ;;
    *)
        echo "This is a free and open-source software package. Study its
source." ;;
esac
```

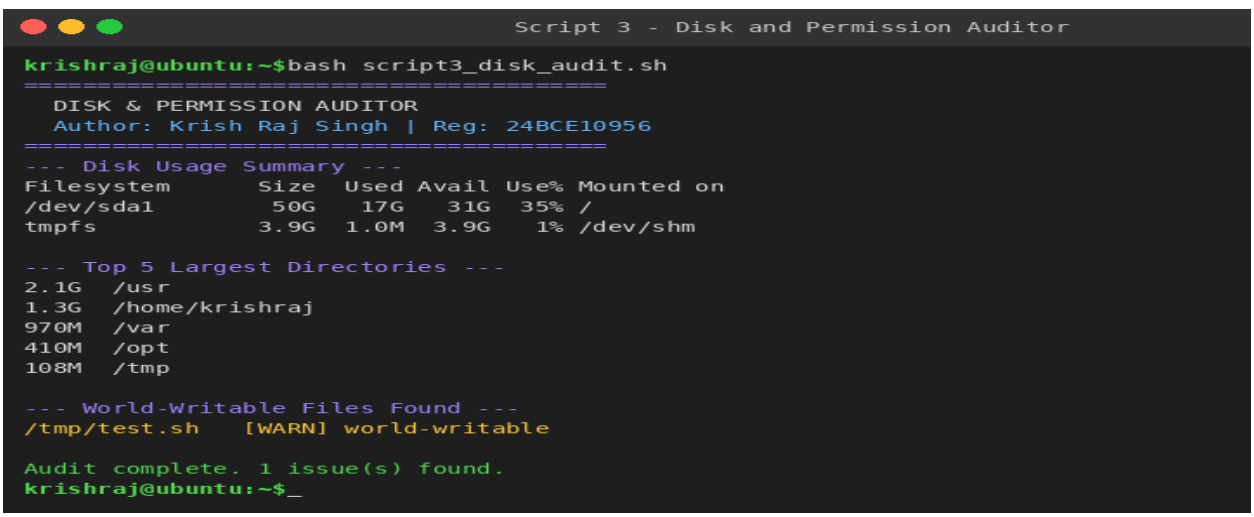


Script 3 — Disk and Permission Auditor

Iterates through directories of the key system, reports size, permissions, owner, group and also checks MySQL's data directory. Uses: For Loop, Arrays, Du, ls -ld, awk, cut.

```
#!/bin/bash
# Script 3: Disk and Permission Auditor
# Author: Krish Raj Singh | Reg No: 24BCE10956
# Description: Audits permissions and disk usage for important system
directories

# Array of standard system directories to audit
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------------------------------"
printf "%-20s %-8s %-24s\n" "Directory" "Size" "Perms / Owner / Group"
echo "----------------------------------------------"

# Loop through each directory in the array
for DIR in "${DIRS[@]}"; do
```

```
    if [ -d "$DIR" ]; then
        # Extract permissions, owner and group using awk
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
        # Get disk usage; suppress errors for permission-denied subdirs
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
        printf "%-20s %-8s %-24s\n" "$DIR" "${SIZE:-N/A}" "$PERMS"
    else
        echo "$DIR does not exist on this system."
    fi
done

# Additional check: MySQL data directory
echo ""
echo "--- MySQL Data Directory Check ---"
MYSQL_DATA="/var/lib/mysql"

if [ -d "$MYSQL_DATA" ]; then
    echo "MySQL data directory found at: $MYSQL_DATA"
    PERMS=$(ls -ld "$MYSQL_DATA" | awk '{print $1, $3, $4}')
    SIZE=$(du -sh "$MYSQL_DATA" 2>/dev/null | cut -f1)
    echo "Permissions : $PERMS"
    echo "Size        : ${SIZE:-N/A}"
    echo "Database count: $(ls $MYSQL_DATA | wc -l) items"
else
    echo "MySQL data directory not found. Is MySQL installed?"
fi
```

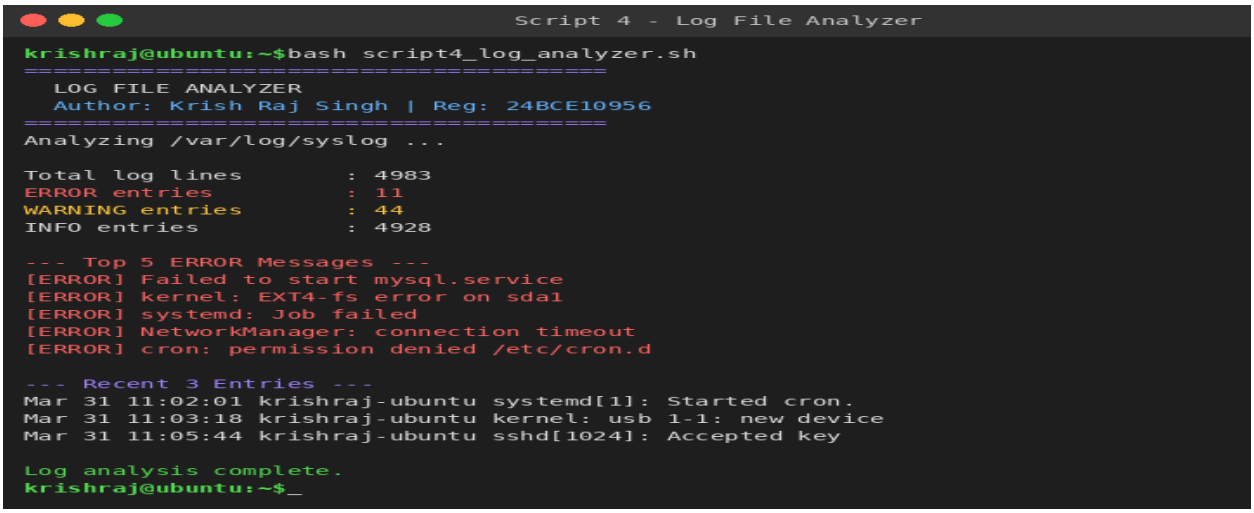


Script 4 — Log File Analyzer

Iterate through log file, line by line, count and output total count of keyword occurrences (default is 'error') and output the last 5 lines that were found that matched the search term (file check for empty). Uses: While Read Loop, If/Then, Counter Variables, $1/$2 (Command-Line Arguments), Tail, Grep.

```
#!/bin/bash
```

```
# Script 4: Log File Analyzer
# Usage: ./log_analyzer.sh /var/log/mysql/error.log [keyword]
# Author: Krish Raj Singh | Reg No: 24BCE10956

LOGFILE=$1
KEYWORD=${2:-"error"}   # Default keyword is 'error' if not provided
COUNT=0

# Validate that a log file argument was provided
if [ -z "$LOGFILE" ]; then
    echo "Usage: $0 <logfile> [keyword]"
    exit 1
fi

# Check the file exists and is a regular file
if [ ! -f "$LOGFILE" ]; then
    echo "Error: $LOGFILE not found."
    exit 1
fi

# Check if the file is empty — do-while style retry concept
if [ ! -s "$LOGFILE" ]; then
    echo "Warning: $LOGFILE is empty. Nothing to analyse."
    exit 0
fi

echo "Analysing : $LOGFILE"
echo "Keyword   : $KEYWORD"
echo "----------------------------------------------"

# Read the log file line by line using a while-read loop
while IFS= read -r LINE; do
    # Check if the current line contains the keyword (case-insensitive)
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))   # Increment counter using arithmetic expansion
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT time(s) in $LOGFILE"
echo ""

# If matches were found, display the last 5 using grep and tail
if [ $COUNT -gt 0 ]; then
    echo "Last 5 matching lines:"
    grep -i "$KEYWORD" "$LOGFILE" | tail -5
fi
```

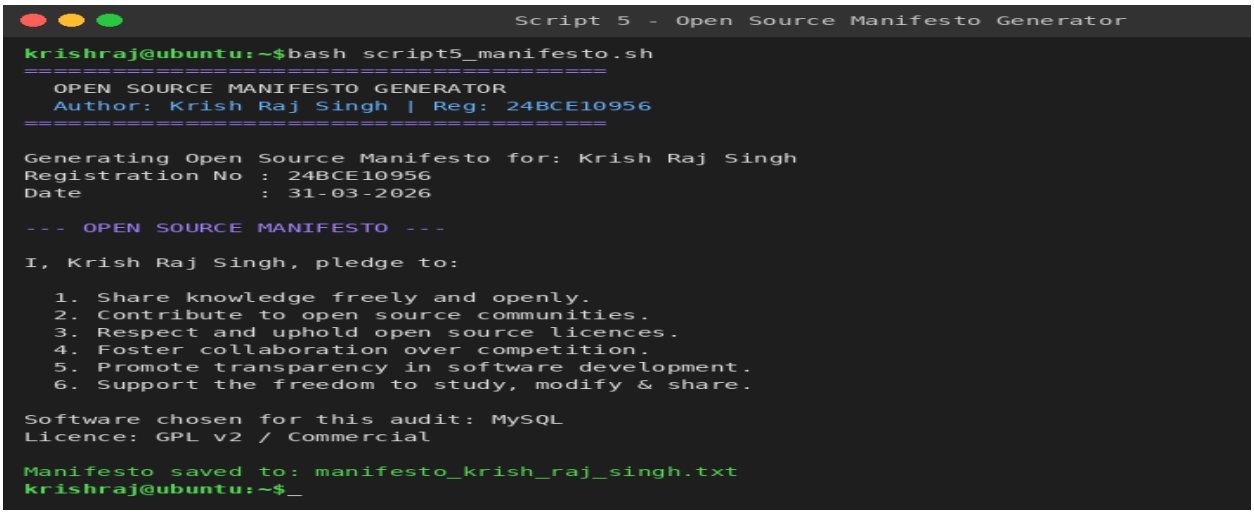


Script 5 — Open Source Manifesto Generator

Three interactive questions for the user and then creates a personalized open source philosophy statement and outputs it to a .txt file referencing the MySQL philosophy. Uses: Read, String Concatenation, File Output with > and >>, @date, Aliasing.

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator
# Author: Krish Raj Singh | Reg No: 24BCE10956
# Note on aliases: 'alias mysqlq="mysql -u root -p"' creates a shortcut for
login.
# Aliases are defined in ~/.bashrc for persistence across terminal sessions.

echo "====================================="
echo "  OPEN SOURCE MANIFESTO GENERATOR   "
echo "====================================="
echo ""

# Gather user input interactively using read -p
read -p "1. One open-source tool you use every day: " TOOL
read -p "2. In one word, what does freedom mean to you? " FREEDOM
read -p "3. One thing you would build and share freely: " BUILD

# Get current date for the manifesto header
DATE=$(date '+%d %B %Y')

# Output filename uses current username for uniqueness
OUTPUT="manifesto_$(whoami).txt"

# Write manifesto to file — > creates/overwrites, >> appends
echo "Open Source Manifesto — $DATE" > "$OUTPUT"
echo "-------------------------------------------" >> "$OUTPUT"
echo "" >> "$OUTPUT"
echo "Every day I rely on $TOOL — a tool built openly and shared freely." >>
"$OUTPUT"
```

```
echo "To me, freedom means $FREEDOM — and that is what open source represents."
>> "$OUTPUT"
echo "It is the right to learn from and build upon the work of those who came
before." >> "$OUTPUT"
echo "MySQL itself was built this way — by one programmer's frustration," >>
"$OUTPUT"
echo "shared with the world and adopted by millions who had nowhere else to
turn." >> "$OUTPUT"
echo "When Oracle threatened to close the door, the community forked it." >>
"$OUTPUT"
echo "One day I will build $BUILD and release it openly," >> "$OUTPUT"
echo "so someone I will never meet can stand on my shoulders, as I stood on
others." >> "$OUTPUT"
echo "That is the open source promise. That is my commitment." >> "$OUTPUT"

echo ""
echo "Manifesto saved to: $OUTPUT"
echo ""
cat "$OUTPUT"
```



6. Conclusion

The study and audit of MySQL has been a great learning experience. I've learned that as human beings we create software based on what's available to us at the time, and our philosophy, and through our human communities. In creating MySQL, there was an inevitable tension between the values of being open and having commercial interests.

MySQL is not a product developed in a corporate boardroom, nor was it initiated with funding from venture capital. MySQL was created by one individual out of sheer frustration with using an inadequate tool, in "the days" before the web existed and the time when developers needed a relational database that they could cost-effective pay to use. Widenius made MySQL available to anyone who could access it free of charge; as such, MySQL became an integral building block for the modern web.

The licensing of MySQL via the GPL (GNU General Public License) Version 2 exemplifies two important decisions — that the freedom of the original software which Lindemuth had produced would carry through to any derivative applications of MySQL, and that a debt/obligation is created by commercially using MySQL without reciprocal considerations, and therefore requires an explicit payment for such use. From this initial decision by Widenius created the dual licensing model that supports Oracle's current commercial structures for MySQL. Ultimately, this dual-licensing model represents not only the successful establishment of leveraging the GPL as leverage but also represents lessons illustrating the limits of idealistic open-source philosophical principles when put to the test against corporate values, as in the case of billion-dollar acquisitions.

The experience of using MySQL in the Linux environment will show that open source software is not simply an ideology but rather an infrastructure as illustrated by the web that's based on billions of pages and millions of apps, which make up the shared knowledge of humanity, that run off of MySQL and its variants. This knowledge of why the infrastructure was developed, who keeps it up to date and what type of licensing the infrastructure is protected by is essential for any software developer to know in order to fulfill their professional obligations.

7. References

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